# Janghyun **Noh**

AI Researcher

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## Education

### Myongji University — *Seoul, S.Korea*

M.S. in Artificial Intelligence — *Sep. 2024 - Aug. 2026*

- Conducted research at the Artificial Intelligence and Data Analytics Laboratory under the supervision of Prof. Minsik Oh.
- Conducted collaborative research with external institutions including the Korea Disease Control and Prevention Agency (KDCA), Seoul National University Hospital (SNUH), Kyungpook National University, and AIgenDrug Co., Ltd.
- Served as a teaching assistant for undergraduate courses including Deep Learning, Machine Learning, Data Mining, and Basic Programming (Fall 2024 – Spring 2026).
- Received the Research Honor Award from the President of Myongji University upon graduation.

### Myongji University — *Seoul, S.Korea*

B.S. in Data Technology, School of Convergence Software — *Mar. 2018 - Aug. 2024*

- Conducted undergraduate research at the Artificial Intelligence and Data Analytics Laboratory under the supervision of Prof. Minsik Oh.
- Conducted collaborative research on clinical and genetic factors associated with hypertension progression in patients with hyperlipidemia in collaboration with the Korea Disease Control and Prevention Agency (KDCA).

### RED School – Alternative School — *Geumsan, S.Korea*

Student — *Mar. 2012 - Jan. 2018*

- Received academic excellence scholarships for two consecutive years.

## Experience

### Artificial Intelligence and Data Analytics Lab, Myongji University — *Seoul, S.Korea*

Graduate Researcher — *Sep. 2024 - Aug. 2026*

- Developed scCap, an annotation-free phenotype prediction framework for single-cell RNA-seq data (first-author paper in Briefings in Bioinformatics), released with code and a Docker image.
- Built scRNA-seq models for MSI/MSS phenotype prediction and applied them to AI-based target discovery (with AIgenDrug).
- Conducted spatial transcriptomics analyses of TNBC Xenium datasets for cellular niche and tumor microenvironment characterization (with SNUH).
- Compared cirrhotic and non-cirrhotic HCC tissues using GeoMx spatial transcriptomics for immunomodulatory target discovery.
- Built a Biomni-based AI agent environment for analyzing CRISPR-Cas9-induced large DNA deletions.
- Developed multimodal COVID-19 severity prediction models combining MIMIC-IV clinical data and MIMIC-CXR chest X-rays (with KDCA).
- Implemented an nf-core/sarek variant-calling pipeline for apple whole-genome sequencing data (with Kyungpook National University).

### Artificial Intelligence and Data Analytics Lab, Myongji University — *Seoul, S.Korea*

Undergraduate Research Intern — *Dec. 2022 - Aug. 2024*

- Developed in-hospital mortality prediction models for ARDS patients using MIMIC-III data, comparing machine learning models across missing-data handling strategies (KSC 2023).
- Designed measurement-based labels and prediction models for hypertension progression in KoGES cohort data (with KDCA).

### Bigleader — *Seoul, S.Korea*

AI Intern (Youth AI Internship) — *Jul. 2022 - Oct. 2022*

- Developed a customs notice analysis and monitoring system for KOTRA.
- Received the KOTRA President's Merit Award and the TriBIG Project Scholarship.

## Publications & Presentations

### Journal Articles

**Annotation-free phenotype prediction using knowledge-augmented clustering from single-cell RNA sequencing data** — *Briefings in Bioinformatics (BIB)*

Published — *2026*

**J. Noh**, Y. Shin, M. Kim, and M. Oh

**Ferroptosis-based therapeutic strategy targeting iron metabolism and SLC7A11 overcomes oxaliplatin resistance in patient-derived pancreatic cancer organoids** — *Journal of Experimental & Clinical Cancer Research (JECCR)*

Published — *2026*

J. Choi, K. Lee, M. Oh, **J. Noh**, Y. Kim, E. Hwang, I. Cho, W. Paik, J. Ryu, J. Ku, and S. Lee

### Conference Presentations

**Patient-Level Metacell Construction with Pre-trained Embeddings for Phenotype Prediction from scRNA-seq** — *KIISE Korea Computer Congress (KCC)* 2026
Oral Presentation
Y. Shin, **J. Noh**, and M. Oh

**Lightweight Analysis of Patient Phenotype Prediction Based on Single-Cell RNA-seq Data Reduction Techniques** — *KIISE Korea Software Congress (KSC)* 2025
Poster Presentation (Best Poster Award)
Y. Shin, **J. Noh**, and M. Oh

**An annotation-free framework for phenotype prediction using hierarchical multiple instance learning with single-cell RNA-sequencing data** — *International Conference on Genome Informatics ISCB-Asia* 2025
Poster Presentation
**J. Noh**, Y. Shin, M. Kim, and M. Oh

**Patient Phenotype Prediction from Single-cell RNA-seq Data through Annotation-free Multiple Instance Learning** — *BIOINFO 2025, Korean Society for Bioinformatics (KSBI)* 2025
Poster Presentation (Excellent Poster Award)
**J. Noh**, Y. Shin, M. Kim, and M. Oh

**AMIL: Automated Cell Grouping Framework for Phenotype Prediction with Multiple Instance Learning** — *KIISE Korea Computer Congress (KCC)* 2025
Oral Presentation (Best Presentation Award)
**J. Noh**, M. Kim, and M. Oh

**Predicting In-Hospital Mortality in Patients with Acute Respiratory Distress Syndrome Using MIMIC-III Data: A Comparative Study of Machine Learning Models with Varied Missing Data Handling Strategies** — *KIISE Korea Software Congress (KSC)* 2023
Poster Presentation
**J. Noh**, G. Choi, and M. Oh

## Honors & Awards

### Awards

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| 2026 | **Research Honor Award**, Myongji University | *Seoul, S.Korea* |
| 2025 | **Best Poster Award**, KIISE Korea Software Congress (KSC) | *Seoul, S.Korea* |
| 2025 | **Excellent Poster Award**, BIOINFO 2025, Korean Society for Bioinformatics (KSBI) | *Seoul, S.Korea* |
| 2025 | **Best Presentation Award**, KIISE Korea Computer Congress (KCC) | *Seoul, S.Korea* |
| 2022 | **President's Merit Award**, Korea Trade-Investment Promotion Agency (KOTRA) | *Seoul, S.Korea* |

### Scholarships

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| 2022 | **TriBIG Project Scholarship**, Korea Trade-Investment Promotion Agency (KOTRA) | *Seoul, S.Korea* |
| 2019 | **Globalization Scholarship**, Myongji University | *Seoul, S.Korea* |
| 2018 | **Scholarship III**, Myongji University | *Seoul, S.Korea* |
| 2016 | **Academic Excellence Scholarship**, RED School | *Geumsan, S.Korea* |
| 2015 | **Academic Excellence Scholarship**, RED School | *Geumsan, S.Korea* |

## Projects

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|---|---|
| **2024 – 2026** | AI-based target discovery platform for CAR-Treg cell therapy in chronic inflammatory disease |
| **2024 – 2026** | Genetic and physiological mechanisms of apple blossom dormancy release under climate change |
| **2024 – 2026** | Spatial transcriptomics-based target discovery and validation for refractory solid tumors |
| **2025 – 2026** | Client-side web platform for detecting large CRISPR-Cas9-induced DNA deletions |
| **2025 – 2026** | GeoMx spatial transcriptomics of cirrhotic vs. non-cirrhotic HCC tissue |
| **2025** | COVID-19 severity prediction using multimodal clinical and imaging data |
| **2023 – 2024** | Hypertension-progression risk groups in hyperlipidemia patients (KoGES cohort) |
| **2023** | CatchYou: criminal composite sketch generation from witness descriptions |
| **2022** | Customs notice analysis and monitoring system for global supply-chain crises |

## Technical Skills

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| **Programming** | Python, R |
| **Frameworks** | PyTorch, TensorFlow, Scikit-learn |
| **Tools** | Linux, Git, Docker, Nextflow |
| **Research** | Machine Learning, Deep Learning, Multimodal Learning |
| **Languages** | Korean (Native), English (Fluent) |
| **Certifications** | Engineer Information Processing (2023), SQL Developer – SQLD (2023) |

## Extracurricular Activities

### Domestic

**Google Developer Student Clubs, Myongji University** — *Seoul, S.Korea*

AI Team Member — *Oct. 2022 - Dec. 2023*

- Engaged in machine learning and deep learning study groups
- Collaborated on AI projects and participated in AI competitions

**Data Youth Campus, MSIT & KDATA** — *Seoul, S.Korea*

Participant (Dongguk University) — *Jul. 2022 - Aug. 2022*

- Completed a hands-on data talent program run by the Ministry of Science and ICT and KDATA, hosted at Dongguk University
- Conducted practical machine learning and data analysis projects

### International

**Camino de Santiago (French Way)** — *Spain*

Pilgrim — *Sep. 2026 - Oct. 2026*

**Medical Mission Trip to the Philippines (GMMA)** — *Manila, Philippines*

Volunteer — *Jan. 2024*

- Participated in medical volunteer activities with international healthcare professionals

**International Work Camp in Iceland** — *Reykjavik, Iceland*

Volunteer — *Dec. 2019 - Jan. 2020*

- Participated in environmental campaigns and intercultural exchange with international volunteers

**Mission Trip to Vietnam** — *Hanoi, Vietnam*

Volunteer — *Aug. 2018*

- Participated in intercultural exchange and community volunteer activities